Biomedical Signal and Image Processing Assignment 1 Report

Github URL https://github.com/scholar-lab/BML735_assignment1
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Entry No. 2025EEZ8518
Course Biomedical Signal and Image Processing(BML7350)

Introduction

The results of the programming exercise given in the assignment is provided including the code and the following explanation is provided.

1. (3.) Inferences from histogram plot of the given image.
2. (7.) Discussion on the results of the Ideal High pass filter and Butterworth Filter.
3. (8.) Discussion on the histogram of the difference image.

1 SNR of the image in gray matter and white matter regions

SNR of the white matter and gray matter is 71.87 and 80.88 respectively. The ROI of the white matter and gray matter is highlighted in green and red box as shown in Fig1.

1.1 SNR of Gray Matter and White Matter

The Signal-to-Noise Ratio (SNR) is used to quantify the quality of an MRI image and is defined as

$$\text{SNR} = \frac{\mu_{\text{ROI}}}{\sigma_{\text{background}}}$$

where μ_{ROI} is the mean signal intensity within the selected tissue region (gray matter or white matter) denoted by red, green box and $\sigma_{\text{background}}$ is the standard deviation of intensities in a background region containing only noise.

The background ROI was selected outside the head region to ensure that only noise was present. The ROIs were chosen with equal sizes (50 px, 50 px) to enable a fair comparison.

The SNR of gray matter is higher than that of white matter because in T2-weighted MRI images gray matter has a longer T2 relaxation time and therefore appears brighter. Since SNR is proportional to the mean signal intensity, the brighter tissue (gray matter) produces a higher SNR than white matter. Hence, SNR measurement confirms the contrast characteristics of tissues in T2-weighted brain MRI.

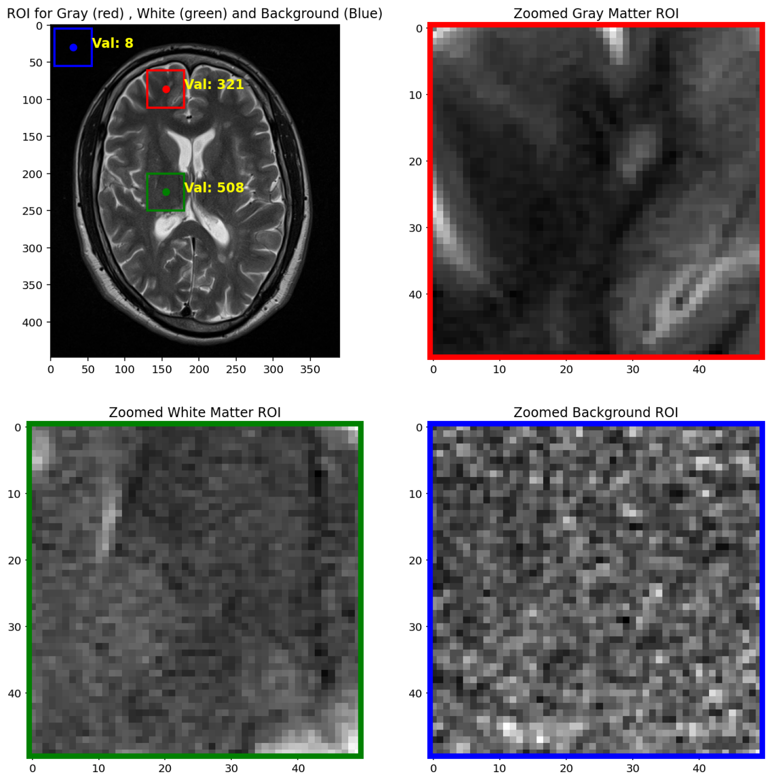

Figure 1: ROI plot on the original image and enlarged view of the ROI for white matter, gray matter and background.

2 Adjusting of intensity value for visualizing background noise

The original image and image showing noise is as shown in Fig 2.

2.1 Intensity Windowing for Noise Visualization

The image intensity range was adjusted to display only the lowest 10% of intensity values. In MRI images, regions outside the skull contain no anatomical signal and therefore consist primarily of acquisition noise.

By restricting the display to low intensities, anatomical structures disappear and only background fluctuations become visible. This allows visualization of the spatial distribution of noise in the image and confirms that the observed variations correspond to background noise rather than tissue structures.

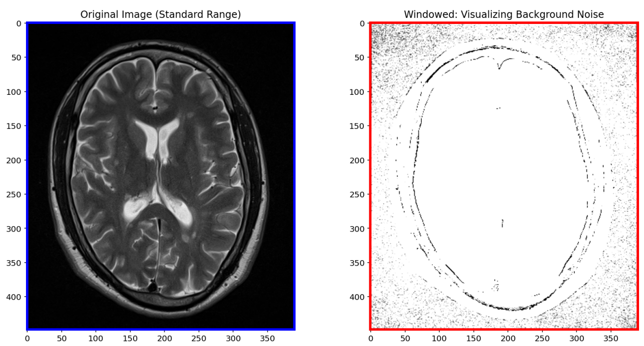

Figure 2: Highlighting bottom 10% of the intensity value to view noise.

3 Plotting Histogram and thresholding to see the lower pixel count intensity values

The original image and its histogram is as shown in Figure 3. A threshold of 2500 is used to limit the pixel count and see the plot of the lower pixel count intensity values more clearly as shown in Figure 4.

3.1 Histogram Analysis of Background Intensities

The histogram of background pixels shows a sharp peak at low intensity values. This peak corresponds to the air surrounding the head where no true MR signal is present. The variations observed in this region arise from scanner noise.

MRI magnitude images are affected by Rician noise; therefore, background pixel values are not exactly zero but distributed around small positive intensities. Tissue regions occupy higher intensity ranges and form broader histogram distributions.

Thus, the histogram (Figure 4) after the curve shows anatomical tissue signal.

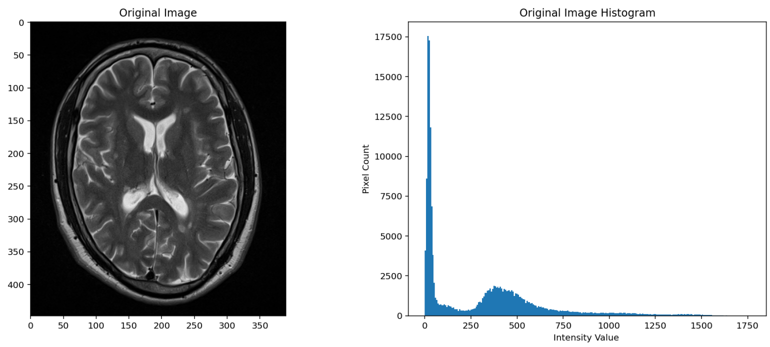

Figure 3: Original image and histogram plot.

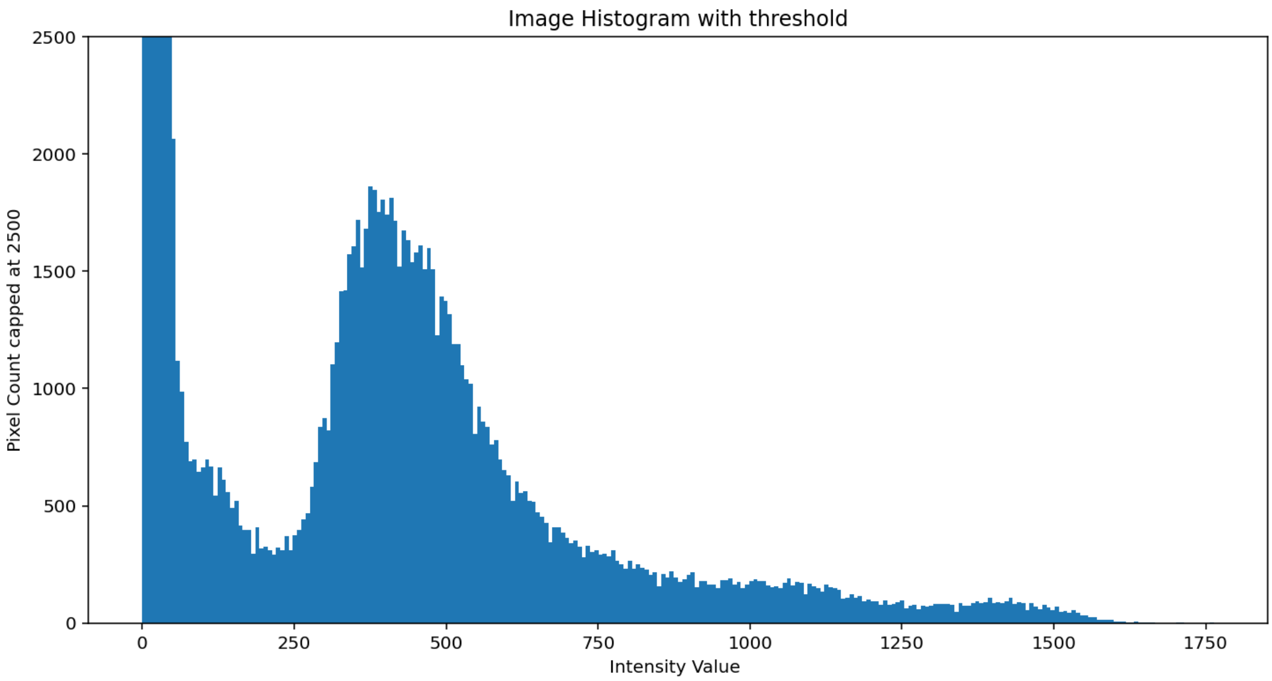

Figure 4: Pixel count threshold at 2500 and histogram plotted to view the lower pixel count intensity values.

4 Application of mean filter and Gaussian filter and comparison of SNR before and after filtering.

The SNR of the original and the filtered image is as shown below:
ORIGINAL:
SNR of Gray Matter: 80.88
SNR of White Matter: 71.87

MEAN FILTER (uniform):
SNR of Gray Matter: 139.24
SNR of White Matter: 123.88

MEAN FILTER (Kernel 2D Convolution):
SNR of Gray Matter: 140.01
SNR of White Matter: 124.59

MEDIAN FILTER (Kernel 2D Convolution):
SNR of Gray Matter: 130.53
SNR of White Matter: 116.42

It is observed that filtering improves the SNR of the image. The plot of the original and the filtered images is as shown in Figure 5.

4.1 Effect of Filtering on SNR

Filtering reduces random intensity fluctuations while preserving the average tissue intensity. Since SNR is defined as the ratio of mean signal to noise standard deviation, reducing noise variance increases the measured SNR.

The mean filter performs local averaging and effectively suppresses Gaussian-like noise present in MRI magnitude images. The Gaussian filter performs weighted averaging and better preserves structural information compared to uniform averaging.

The observed increase in SNR after filtering does not indicate an improvement in scanner signal, but rather a reduction in noise standard deviation due to smoothing.

The median filter is more effective for impulse (salt-and-pepper) noise; however, MRI noise is not impulsive in nature. Therefore, it produces less SNR improvement compared to mean and Gaussian filtering.

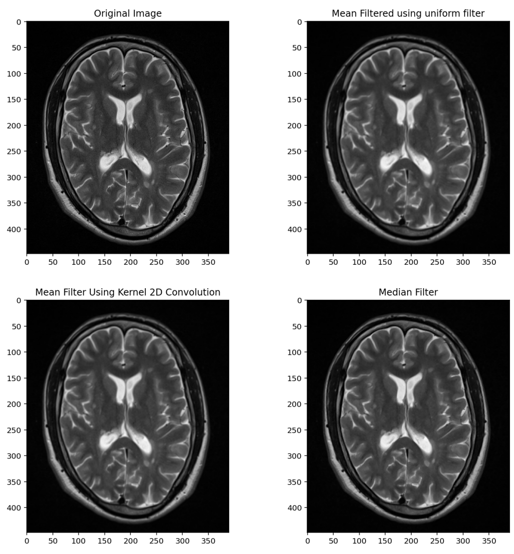

Figure 5: Mean and Median Filter plots.

5 Applying Mean filter using Fourier transform operation

Mean filter is applied using FFT. The original and filtered image is as shown in Figure 6.

5.1 Mean Filtering using Fourier Transform

According to the convolution theorem,

$$f(x,y) * h(x,y) \iff F(u,v)\,H(u,v)$$

i.e., convolution in the spatial domain is equivalent to multiplication in the frequency domain.

Therefore, instead of directly performing spatial convolution with a 3×3 averaging kernel, the Fourier transform of the image was multiplied by the Fourier transform of the filter and then inverse transformed. The resulting image matches the spatial filtering result, confirming the convolution theorem.

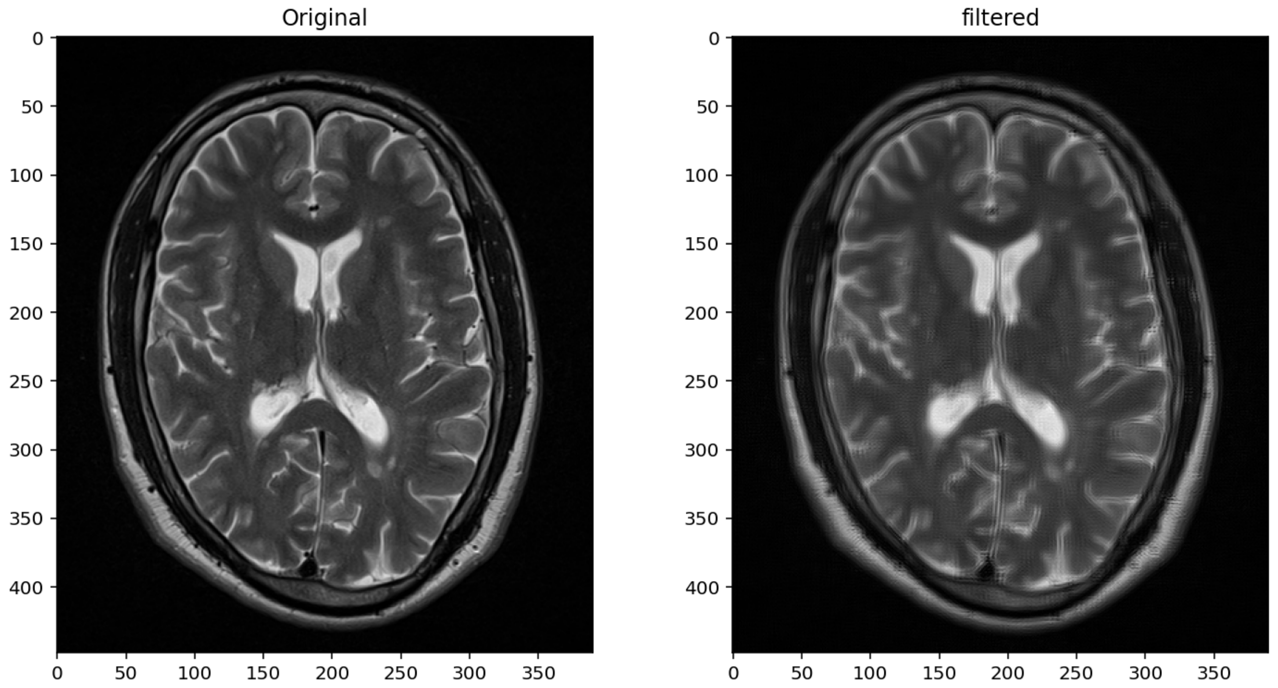

Figure 6: Mean Filter using Fourier Transform.

6 Computation of Fourier Transform of the image

The Fourier transform of the image is performed. The plots of the original image, Log scale Magnitude, Phase and Linear scale magnitude is as shown in Figure 7.

6.1 Fourier Transform Analysis

The magnitude spectrum shows a bright region at the center representing low spatial frequencies. These frequencies correspond to slowly varying components such as overall illumination and large anatomical structures.

The outer regions of the spectrum contain high spatial frequencies corresponding to edges and fine details. The phase spectrum appears random but is essential for structural reconstruction. If phase information is removed, the image structure cannot be reconstructed even when magnitude is preserved.

Thus, the magnitude represents intensity distribution, while the phase carries most of the structural information of the image.

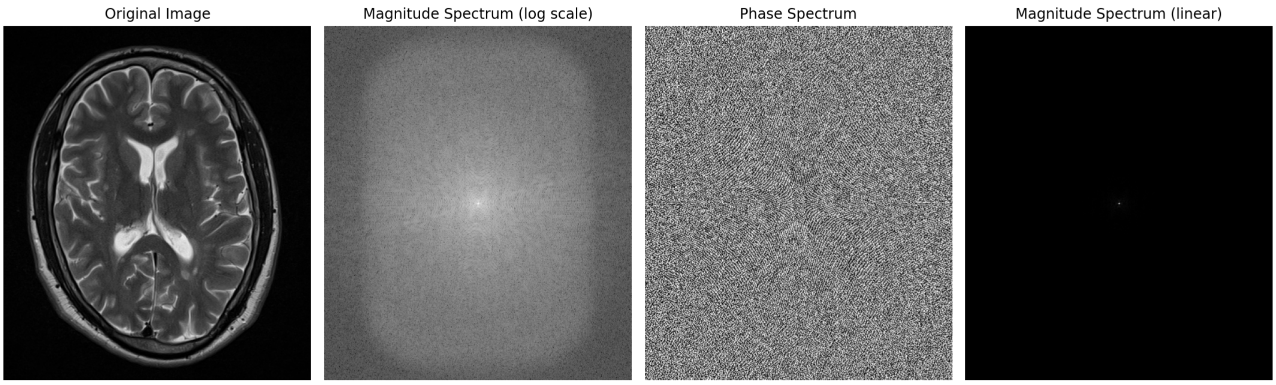

Figure 7: Fourier Transform - Magnitude and Phase plot

7 Application of high pass filters and discussion of the results

The result of the high pass ideal filter and Butterworth filter is as shown in Figure 8.

7.1 Comparison of Ideal and Butterworth High-Pass Filters

High-pass filtering suppresses low spatial frequencies and enhances edges and fine structures.

The Ideal High-Pass Filter uses an abrupt cutoff in the frequency domain. This sharp discontinuity produces oscillations in the spatial domain known as the Gibbs phenomenon, which appears as bright and dark halos around edges.

The Butterworth High-Pass Filter provides a gradual transition between passband and stopband. Because its frequency response is smooth, spatial oscillations are reduced and edge enhancement appears more natural.

Although the ideal filter removes background more strongly, it introduces noticeable artifacts. Therefore, the Butterworth filter is more suitable for medical image processing since it enhances edges while preserving anatomical boundaries without significant ringing artifacts.

Figure 8: High pass filter plot

8 Computation and histogram of difference image between the original image and filtered image. Discussion on the histogram.

The difference image and its histogram plot is as shown in Figure 9.

8.1 Discussion on Difference Image Histogram

The difference image represents the high-frequency components removed by the mean filter. These components include both noise and fine anatomical details such as edges and texture.

Therefore, the difference image does not represent pure noise. The histogram indicates that most removed components correspond to low-intensity fluctuations (noise), while a smaller portion corresponds to structural information.

This demonstrates that the mean filter suppresses high spatial frequencies, improving smoothness but slightly reducing edge sharpness.

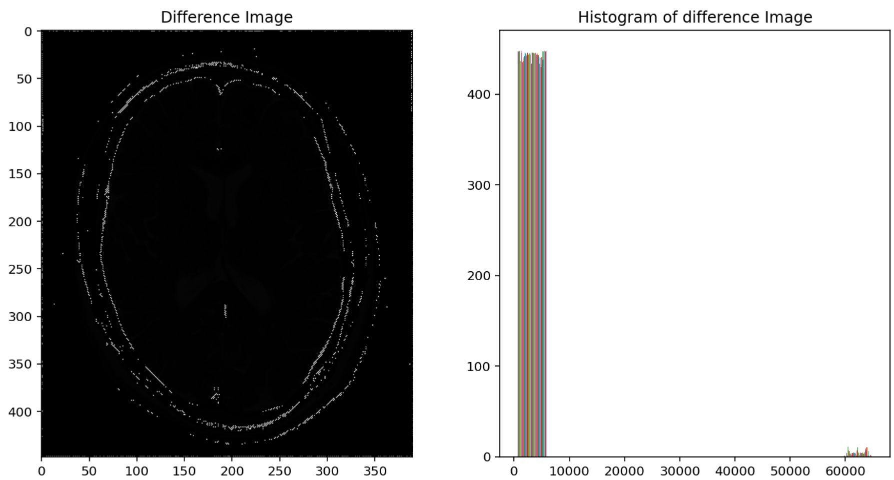

Figure 9: Difference Image and Histogram plot